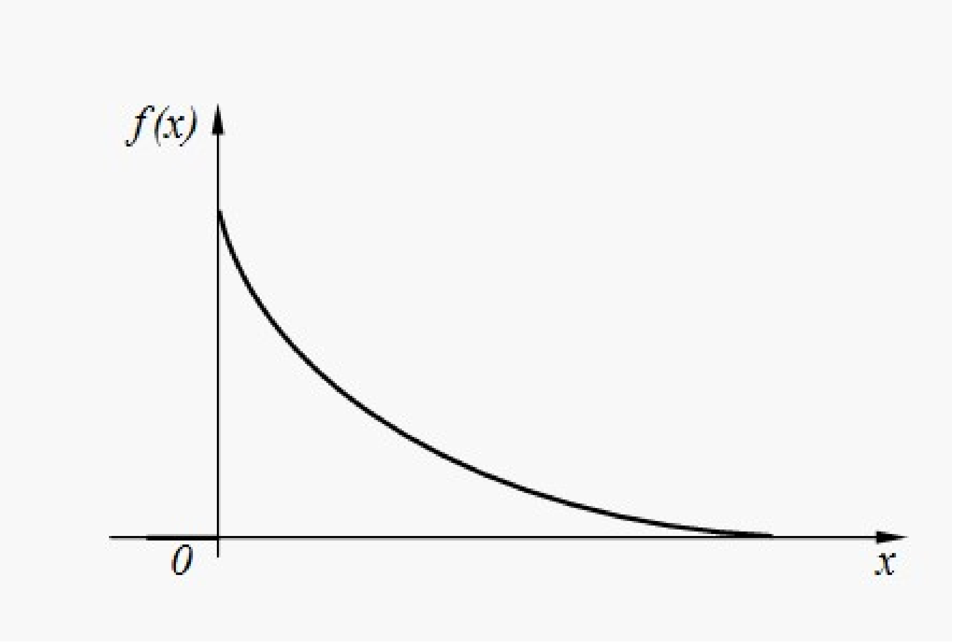
2.3.3 Exponent distribution.

Continuous random variable X has an *exponent distribution* if its density function is expressed by the formula:

$$f(x) = \begin{cases} \lambda \cdot e^{-\lambda \cdot x} & at\ x > 0 \\ 0 & at\ x < 0 \end{cases} \qquad (31)$$

Graph of density function (31) is represented on Fig.

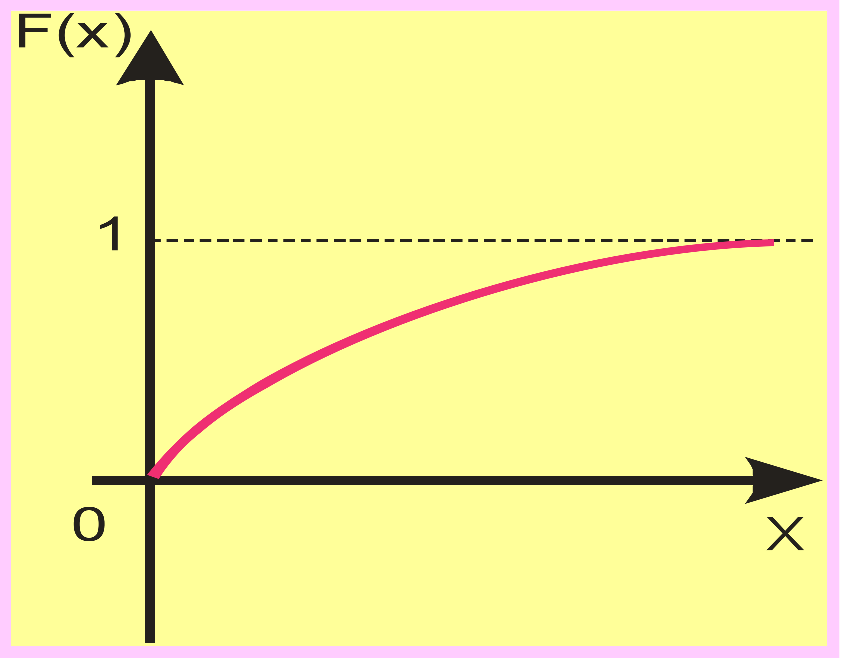


Distribution function of *exponent distribution* law $F(x)$ is:

$$F(x) = P\{X < x\} = \int_{-\infty}^{x} f(x)\,dx$$

$$= \int_{0}^{x} \lambda \cdot e^{-\lambda \cdot x}\,dx = 1 - e^{-\lambda \cdot x}$$

Graph of **d*istribution function*** is represented on Fig.



Time T of non-failure operation of computer system is a random variable having exponent distribution with parameter λ, which physical sense – an average number of refusals in unit of time.

T is the average operating time of a computer or any other device.

Mathematical expectation: $E(X) = \mu_X = \dfrac{1}{\lambda} = T,$

Dispersion: $D(X) = \dfrac{1}{\lambda}$

Standard deviation: $\sigma = \sqrt{\dfrac{1}{\lambda}}$

$\lambda = \frac{1}{10}$

Example 2.7

Random variable X - the operating time of the refrigerator has an exponential distribution. Find the probability that the refrigerator will work for at least 20 years if the average operating time of the refrigerator is 10 years?

SOLUTION $T = \frac{1}{\lambda} = 10$ $\lambda = \frac{1}{10}$

Using the distribution function, we find the probability that the refrigerator will work for less than 20 years

$$F(x) = P\{X < x\} = 1 - e^{-\lambda \cdot x}$$

$$F(x) = P\{X < 20\} = F(20) = 1 - e^{-\frac{20}{10}}$$

The probability that the refrigerator will work for more than 20 years is found as the probability of the opposite event

$$P(X \geq 20) = 1 - P(X < 20) = 1 - F(20) =$$

$$= 1 - (1 - e^{-\frac{20}{10}}) = e^{-2} \approx 0{,}135$$

Example 2.8

The time for the repair of the device has an exponential distribution law with an average time of 5 hours.

What is the probability of a repair in less than two hours?

What is the probability of repair more than 3 hours?